

derivative bounded away from zero and infinity; (v) β_0 is an interior point of a compact set $\Theta \subset \mathbb{R}^k$; (vi) $E_M Y^2 < \infty$, $\|X_k\|$ is bounded, and $E[X_k X'_k | D = 1]$ is non-singular.

Under this assumption, γ_0 can be estimated by maximum likelihood:

$$\hat{\gamma} = \arg \max_{\gamma \in \Gamma} \frac{1}{n} \sum_{i=1}^n D_i \log \pi(X'_i \gamma) + (1 - D_i) \log(1 - \pi(X'_i \gamma)).$$

Then, $\hat{\pi}(X_i) = \pi(X'_i \hat{\gamma})$. Boundedness of X is not a necessary restriction but allows Assumption 4.2 to encompass maximum likelihood estimation of the logit, probit and linear probability models. For simplicity, and since in most cases X_k is some subset of the variables in X , X_k is also assumed to be bounded. If the covariates are discrete and the vector X is saturated with indicators of all the possible values of the covariates, then the functional form of π is not restrictive and the estimation of $\hat{\pi}(X)$ is completely non-parametric.

Let $\dot{\pi} = \partial \pi(v) / \partial v$ and $\dot{\pi}_0 = \dot{\pi}(X' \gamma_0)$. Under standard regularity conditions (e.g. Assumption 4.2(i)–(iv)), $\hat{\gamma}$ is asymptotically linear, that is $n^{1/2}(\hat{\gamma} - \gamma_0) = n^{-1/2} \sum_{i=1}^n \psi_{\gamma_0}(Z_i) + o_p(1)$, where

$$\psi_{\gamma_0}(Z) = E \left[\frac{\dot{\pi}_0^2}{\pi_0(1 - \pi_0)} X X' \right]^{-1} X \frac{\dot{\pi}_0}{\pi_0(1 - \pi_0)} (D - \pi_0). \quad (17)$$

Let

$$M_{\gamma_0} = E_M \left[X_k \left(\frac{T - \lambda}{\lambda(1 - \lambda)} \frac{D - 1}{(1 - \pi_0)^2} Y - X'_k \beta_0 \right) \dot{\pi}_0 X' \right].$$

Theorem 4.3. *If $n_0, n_1 \rightarrow \infty$, $n_1/(n_0 + n_1) = \lambda \in (0, 1)$ and Assumptions 3.1, 3.3 and 4.2 hold, then $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, V)$, where $V = Q^{-1} \Sigma Q^{-1}$, $Q = E[X_k D X'_k]$, $\Sigma = E_M[\psi \psi']$, $\psi = m(Z, \beta_0, \gamma_0) + M_{\gamma_0} \psi_{\gamma_0}$, $m(Z, \beta_0, \gamma_0) = X_k \pi_0 [\varphi_0 \cdot Y - X'_k \beta_0]$.*

Let $\hat{V} = \hat{Q}^{-1} \hat{\Sigma} \hat{Q}^{-1}$, where

$$\begin{aligned} \hat{Q} &= \frac{1}{n} \sum_{i=1}^n X_{ki} D_i X'_{ki}, \quad \hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i \hat{\psi}'_i, \\ \hat{M}_{\hat{\gamma}} &= \frac{1}{n} \sum_{i=1}^n X_{ki} \left(\frac{T_i - \lambda}{\lambda(1 - \lambda)} \frac{D_i - 1}{(1 - \hat{\pi}(X_i))^2} Y_i - X'_{ki} \hat{\beta} \right) \dot{\pi}(X'_i \hat{\gamma}) X'_i, \\ \hat{\psi}_{\hat{\gamma}}(Z_i) &= \left(\frac{1}{n} \sum_{i=1}^n \frac{\dot{\pi}(X'_i \hat{\gamma})^2}{\hat{\pi}(X_i)(1 - \hat{\pi}(X_i))} X_i X'_i \right)^{-1} X_i \frac{\dot{\pi}(X'_i \hat{\gamma})}{\hat{\pi}(X_i)(1 - \hat{\pi}(X_i))} (D_i - \hat{\pi}(X_i)), \end{aligned}$$

and $\hat{\psi}_i = X_{ki} \hat{\pi}(X_i) (\hat{\varphi}_i Y_i - X'_{ki} \hat{\beta}) + \hat{M}_{\hat{\gamma}} \hat{\psi}_{\hat{\gamma}}(Z_i)$. The following theorem establishes consistency of $\hat{V}$.

Theorem 4.4. *If the assumptions of Theorem 4.3 hold and $\pi(v)$ is twice differentiable with bounded second derivative in $\mathcal{V}$, then $\hat{V} \xrightarrow{p} V$.*

5. CONCLUSIONS

In this article, I have introduced a family of semiparametric difference-in-differences estimators of treatment effects based on conditional identification restrictions. These estimators may be particularly appropriate when the distribution of observed characteristics that are thought to be related to the dynamics of the outcome variable differs between treated and untreated.

Identification does not entail parametric restrictions. However, the methods presented here can be used to estimate parsimonious parametric approximations to conditional versions of the average treatment effect for the treated.

APPENDIX. PROOFS

Proof of Lemma 3.1. For $0 < P(D = 1 | X) < 1$, we have that

$$\begin{aligned} E[\rho_0 \cdot (Y(1) - Y(0)) | X] &= E[\rho_0 \cdot (Y(1) - Y(0)) | X, D = 1] \cdot P(D = 1 | X) \\ &\quad + E[\rho_0 \cdot (Y(1) - Y(0)) | X, D = 0] \cdot P(D = 0 | X) \\ &= E[Y(1) - Y(0) | X, D = 1] - E[Y(1) - Y(0) | X, D = 0]. \end{aligned}$$

Applying equation (9) we obtain the result. $\parallel$

Proof of Proposition 3.1. Let $G(\theta) = E[P(D = 1 | X) \cdot \{\rho_0 \cdot (Y(1) - Y(0)) - g(X_k; \theta)\}^2]$. Adding and subtracting $E[Y^1(1) - Y^0(1) | X_k, D = 1]$, we obtain

$$\begin{aligned} G(\theta) &= E[P(D = 1 | X) \cdot \{\rho_0 \cdot (Y(1) - Y(0)) - E[Y^1(1) - Y^0(1) | X_k, D = 1]\}^2] \\ &\quad + E[P(D = 1 | X) \cdot \{E[Y^1(1) - Y^0(1) | X_k, D = 1] - g(X_k; \theta)\}^2] \\ &\quad + 2E[P(D = 1 | X) \cdot (\rho_0 \cdot (Y(1) - Y(0)) - E[Y^1(1) - Y^0(1) | X_k, D = 1]) \\ &\quad \times (E[Y^1(1) - Y^0(1) | X_k, D = 1] - g(X_k; \theta))]. \end{aligned} \tag{A.1}$$

The first term on the R.H.S. of equation (A.1) does not depend on θ .

By the law of iterated expectations, the second term on the R.H.S. of equation (A.1) is equal to

$$\begin{aligned} &E[D \cdot \{E[Y^1(1) - Y^0(1) | X_k, D = 1] - g(X_k; \theta)\}^2] \\ &= E[\{E[Y^1(1) - Y^0(1) | X_k, D = 1] - g(X_k; \theta)\}^2 | D = 1]P(D = 1). \end{aligned}$$

Therefore, by Assumption 3.2 and equation (11), the second term on the R.H.S. of equation (A.1) is minimized at θ_0 .

The expectation in the third term on the R.H.S. of equation (A.1) is equal to

$$\begin{aligned} &E[E[P(D = 1 | X) \cdot (\rho_0 \cdot (Y(1) - Y(0)) - E[Y^1(1) - Y^0(1) | X_k, D = 1]) | X_k] \\ &\quad \times (E[Y^1(1) - Y^0(1) | X_k, D = 1] - g(X_k; \theta))]. \end{aligned}$$

Applying the law of iterated expectations and Lemma 3.1:

$$\begin{aligned} &E[P(D = 1 | X) \cdot \rho_0 \cdot (Y(1) - Y(0)) | X_k] = E[P(D = 1 | X)E[\rho_0 \cdot (Y(1) - Y(0)) | X] | X_k] \\ &= E[P(D = 1 | X)E[Y^1(1) - Y^0(1) | X, D = 1] | X_k] = E[E[D(Y^1(1) - Y^0(1)) | X] | X_k] \\ &= E[D(Y^1(1) - Y^0(1)) | X_k] = P(D = 1 | X_k) \cdot E[Y^1(1) - Y^0(1) | X_k, D = 1]. \end{aligned}$$

Therefore,

$$\begin{aligned} &E[P(D = 1 | X) \cdot (\rho_0 \cdot (Y(1) - Y(0)) - E[Y^1(1) - Y^0(1) | X_k, D = 1]) | X_k] \\ &= E[P(D = 1 | X) \cdot \rho_0 \cdot (Y(1) - Y(0)) | X_k] - P(D = 1 | X_k)E[Y^1(1) - Y^0(1) | X_k, D = 1] = 0. \end{aligned}$$

Consequently, the third term on the R.H.S. of equation (A.1) is equal to zero, and the result of the proposition holds. $\parallel$

Proof of Lemma 3.2. Notice that

$$\begin{aligned} E_M[\varphi_0 \cdot Y | X] &= E_M[E_M[\varphi_0 \cdot Y | X, T] | X] = E_M[E[\varphi_0 \cdot Y | X, T] | X] \\ &= E[\rho_0 \cdot Y | X, T = 1] - E[\rho_0 \cdot Y | X, T = 0] = E[\rho_0 \cdot Y(1) | X] - E[\rho_0 \cdot Y(0) | X] \\ &= \{E[Y(1) | X, D = 1] - E[Y(1) | X, D = 0]\} - \{E[Y(0) | X, D = 1] - E[Y(0) | X, D = 0]\}, \end{aligned}$$

and the result follows from equation (9). $\parallel$

Proof of Proposition 3.2. The proof follows the same steps as the proof of Proposition 3.1.

Proof of Theorem 4.1. Let $\|\cdot\|_\infty$ denote the supremum norm. By Assumption 4.1(i), (ii) and Theorem 4 in Newey (1997), it follows that $\|\hat{\pi} - \pi_0\|_\infty = O_P(K \cdot [(K/n)^{1/2} + K^{-s/r}]) = o_P(1)$. Let $m(Z, \beta, \pi) = X_k \pi(\varphi(\pi) \cdot Y - X_k' \beta)$.

Assumption 4.1(v) guarantees that $E_M[m(Z, \beta, \pi_0)]$ has a unique zero at $\beta_0 = (E[X_k \pi_0 X'_k])^{-1} E_M[X_k \pi_0 \varphi_0 Y]$. In addition,

$$\begin{aligned} \sup_{\beta \in \Theta} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \hat{\pi}) - E_M[m(Z, \beta, \pi_0)] \right\| &\leq \sup_{\beta \in \Theta} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \hat{\pi}) - m(Z_i, \beta, \pi_0) \right\| \\ &+ \sup_{\beta \in \Theta} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \pi_0) - E_M[m(Z, \beta, \pi_0)] \right\|. \end{aligned} \quad (\text{A.2})$$

Let

$$\varphi_{0i} = \frac{T_i - \lambda}{\lambda \cdot (1 - \lambda)} \cdot \frac{D_i - \pi_0(X_i)}{\pi_0(X_i) \cdot (1 - \pi_0(X_i))}.$$

Notice that, $m(Z_i, \beta, \hat{\pi}) - m(Z_i, \beta, \pi_0) = X_{ki} \{(\hat{\pi}(X_i) \hat{\varphi}_i - \pi_0(X_i) \varphi_{0i}) Y_i - (\hat{\pi}(X_i) - \pi_0(X_i)) X'_{ki} \beta\}$, and

$$\hat{\pi}(X_i) \hat{\varphi}_i - \pi_0(X_i) \varphi_{0i} = \frac{T_i - \lambda}{\lambda(1 - \lambda)} \cdot \frac{D_i - 1}{(1 - \hat{\pi}(X_i))(1 - \pi_0(X_i))} \cdot (\hat{\pi}(X_i) - \pi_0(X_i)).$$

By Assumption 4.1(iii) and uniform convergence of $\hat{\pi}$, with probability approaching one $\hat{\pi}$ is bounded away from one and there is a constant C such that

$$\sup_{\beta \in \Theta} \left\| \frac{1}{n} \sum_{i=1}^n m(Z_i, \beta, \hat{\pi}) - m(Z_i, \beta, \pi_0) \right\| \leq C \cdot \frac{1}{n} \sum_{i=1}^n \{\|X_{ki} Y_i\| + \|X_{ki}\|^2 \|\beta\|\} |\hat{\pi} - \pi_0|_\infty.$$

Therefore, the first term of the R.H.S. of equation (A.2) is $o_p(1)$. Assumption 4.1 also implies that $\|m(Z, \beta, \pi_0)\|$ is dominated by an integrable function. Then, since $m(Z, \beta, \pi_0)$ is continuous at each $\beta \in \Theta$ compact, Lemma 2.4 in Newey and McFadden (1994) implies that the second term of the R.H.S. of equation (A.2) is also $o_p(1)$ and that $E[m(Z, \beta, \pi_0)]$ is continuous. Now, the usual consistency argument for estimators based on moment equations (e.g. van der Vaart (1998, Theorem 5.9)) proves that $\hat{\beta} \xrightarrow{P} \beta_0$.

Since $|\hat{\pi} - \pi_0|_\infty$ converges in probability to zero, with probability approaching one $n^{-1} \sum_{i=1}^n X_{ki} \hat{\pi}(X_i) X'_{ki}$ is non-singular and

$$n^{1/2}(\hat{\beta} - \beta_0) = \left(\frac{1}{n} \sum_{i=1}^n X_{ki} \hat{\pi}(X_i) X'_{ki} \right)^{-1} \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n X_{ki} \hat{\pi}(X_i) (\hat{\varphi}_i Y_i - X'_{ki} \beta_0).$$

Since X_k has second moments,

$$\frac{1}{n} \sum_{i=1}^n X_{ki} \hat{\pi}(X_i) X'_{ki} = E[X_k D X'_k] + o_p(1).$$

Now, let us prove that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n m(Z_i, \beta_0, \hat{\pi}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_{ki} \hat{\pi}(X_i) (\hat{\varphi}_i Y_i - X'_{ki} \beta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(Z_i) + o_p(1).$$

Consider

$$\Lambda(Z, \pi, \beta, \tilde{\pi}) = \left(\frac{\partial}{\partial \pi} m(Z, \beta, \pi) \Big|_{\pi=\tilde{\pi}} \right) \cdot \pi = X_k \left[\frac{T - \lambda}{\lambda \cdot (1 - \lambda)} \cdot \frac{D - 1}{(1 - \tilde{\pi})^2} \cdot Y - X'_k \beta \right] \cdot \pi.$$

It can be easily seen that for $\pi, \tilde{\pi}$ bounded away from one we have

$$m(Z, \beta, \pi) - m(Z, \beta, \tilde{\pi}) - \Lambda(Z, \pi - \tilde{\pi}, \beta, \tilde{\pi}) = X_k \cdot \frac{T - \lambda}{\lambda \cdot (1 - \lambda)} \cdot \frac{D - 1}{(1 - \pi)(1 - \tilde{\pi})^2} \cdot Y \cdot (\pi - \tilde{\pi})^2.$$

Therefore, since, with probability approaching one, $(1 - \hat{\pi})^{-1}$ is bounded by some constant C , we have

$$\begin{aligned} &\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n m(Z_i, \beta_0, \hat{\pi}) - m(Z_i, \beta_0, \pi_0) - \Lambda(Z_i, \hat{\pi} - \pi_0, \beta_0, \pi_0) \right\| \\ &\leq n^{1/2} |\hat{\pi} - \pi_0|_\infty^2 C \cdot \frac{1}{n} \sum_{i=1}^n \left\| X_{ki} \cdot \frac{T_i - \lambda}{\lambda \cdot (1 - \lambda)} \cdot \frac{D_i - 1}{(1 - \pi_0)^2} \cdot Y_i \right\| = o_p(1). \end{aligned}$$

This result holds because Assumption 4.1(i) implies that $n^{1/2} K^2 \cdot (K/n + K^{-2s/r}) \rightarrow 0$, and therefore $n^{1/2} |\hat{\pi} - \pi_0|_\infty^2 = o_p(1)$. The assumptions $E_M|Y| < \infty$, $\|X_k\|$ bounded and π_0 bounded away from one take care of the sample average term. Therefore

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n m(Z_i, \beta_0, \hat{\pi}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n m(Z_i, \beta_0, \pi_0) + \frac{1}{\sqrt{n}} \sum_{i=1}^n \Lambda(Z_i, \hat{\pi} - \pi_0, \beta_0, \pi_0) + o_p(1).$$

Note that $\|\Lambda(Z, \pi, \beta_0, \pi_0)\| \leq b(Z) \cdot |\pi|_\infty$ with $b(Z) = \|X_k[(T - \lambda)/(\lambda \cdot (1 - \lambda))] \cdot [(D - 1)/(1 - \pi_0)^2] \cdot Y - X'_k \beta_0\|$. By $E_M Y^2 < \infty$ and $\|X_k\|$ bounded, we have that $E_M b(Z)^2 < \infty$. Then, it follows from the proof of Theorem 6.1 in Newey (1994) that

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\Lambda(Z_i, \hat{\pi} - \pi_0, \beta_0, \pi_0) - E_M[\Lambda(Z, \hat{\pi} - \pi_0, \beta_0, \pi_0)]\} \\ &= O_p(K^{-s/r}) + O_p(K \cdot [(K/n)^{1/2} + K^{-s/r}]) = o_p(1). \end{aligned}$$

Note that $E_M[\Lambda(Z, \pi, \beta_0, \pi_0)] = E[\delta(X)\pi(X)]$. If:

- (a) For each K , there exists ξ_K such that $E[\|\delta(X) - \xi_K p^K(X)\|^2] \rightarrow 0$ and $nK^{-2s/r} E[\|\delta(X) - \xi_K p^K(X)\|^2] \rightarrow 0$,
- (b) $K^5/n \rightarrow 0$,
- (c) $K^2 \cdot K^{-2s/r} \rightarrow 0$,

then, Assumption 6.6 in Newey (1994) holds for $\Lambda(Z, \pi, \beta_0, \pi_0)$. Condition (a) holds by Assumption 4.1(i) and by square-integrability of $\delta(X)$. Conditions (b) and (c) follow from Assumption 4.1(i). Under such conditions, Newey (1994) shows that

$$n^{1/2} E_M[\Lambda(Z, \hat{\pi} - \pi_0, \beta_0, \pi_0)] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \delta(X_i) \cdot (D_i - \pi_0(X_i)) + o_p(1).$$

So the result of the theorem follows from existence of second moments of ψ . $\parallel$

Proof of Theorem 4.2. By the law of large numbers $\hat{Q}$ converges in probability to Q , which is non-singular. Notice that

$$\begin{aligned} & X_{ki} \hat{\pi}(X_i) (\hat{\varphi}_i Y_i - X'_{ki} \hat{\beta}) - X_{ki} \pi_0(X_i) (\varphi_{0i} Y_i - X'_{ki} \beta_0) \\ &= X_{ki} \left(\frac{T_i - \lambda}{\lambda(1 - \lambda)} \cdot \frac{D_i - 1}{(1 - \hat{\pi}(X_i))(1 - \pi_0(X_i))} \right) Y_i (\hat{\pi}(X_i) - \pi_0(X_i)) - X_{ki} X'_{ki} (\hat{\beta} - \beta_0). \end{aligned}$$

Now, since $\|X_{ki}\|$ is bounded, $E_M Y^2 < \infty$, π_0 is bounded away from zero, and $|\hat{\pi} - \pi_0|_\infty$ and $\|\hat{\beta} - \beta_0\|$ are $o_p(1)$, we obtain

$$\frac{1}{n} \sum_{i=1}^n \|X_{ki} \hat{\pi}(X_i) (\hat{\varphi}_i Y_i - X'_{ki} \hat{\beta}) - X_{ki} \pi_0(X_i) (\varphi_{0i} Y_i - X'_{ki} \beta_0)\|^2 = o_p(1). \quad (\text{A.3})$$

For π bounded away from one, we have that

$$\Lambda(Z, \tilde{\pi}, \beta, \pi) - \Lambda(Z, \tilde{\pi}, \beta_0, \pi_0) = \left\{ \left(X_k \frac{T - \lambda}{\lambda(1 - \lambda)} \cdot \frac{(1 - \pi_0) + (1 - \pi)}{(1 - \pi)^2(1 - \pi_0)^2} (D - 1) Y \right) (\pi - \pi_0) - X_k X'_k (\beta - \beta_0) \right\} \tilde{\pi}.$$

Therefore, there exists some function, $b(\cdot)$, such that $\|\Lambda(Z, \tilde{\pi}, \beta, \pi) - \Lambda(Z, \tilde{\pi}, \beta_0, \pi_0)\| \leq b(Z)(|\pi - \pi_0|_\infty + \|\beta - \beta_0\|)|\tilde{\pi}|_\infty$ with $E_M b(Z) < \infty$. This result, along with $K^7/n \rightarrow 0$ guarantees that

$$\frac{1}{n} \sum_{i=1}^n \|\hat{\delta}(X_i)(D_i - \hat{\pi}(X_i)) - \delta(X_i)(D_i - \pi_0(X_i))\|^2 = o_p(1) \quad (\text{A.4})$$

(see Newey, 1994). Equations (A.3) and (A.4), along with the Triangle and Hölder's inequalities, imply $\hat{\Sigma} \xrightarrow{P} \Sigma$. As a result, $\hat{V} \xrightarrow{P} V$. $\parallel$

Proof of Theorem 4.3. First, let us prove that under Assumption 4.2(i)–(iv), $\hat{\gamma}$ is asymptotically linear with influence function given by equation (17). By Assumption 4.2(ii), $E[XX']$ is non-singular, therefore $(\gamma - \gamma_0)'E[XX'](\gamma - \gamma_0) > 0$ for $\gamma \neq \gamma_0$. As a result, for $\gamma \neq \gamma_0$, $X'\gamma \neq X'\gamma_0$ in a set of positive probability. Since $\pi(v)$ is strictly increasing in $v \in V$, we have that $\pi(X'\gamma) \neq \pi(X'\gamma_0)$ in a set of positive probability. Let $m_\gamma = D \log(\pi(X'\gamma)) + (1 - D) \log(1 - \pi(X'\gamma))$.

$$\begin{aligned} E[m_\gamma - m_{\gamma_0} | X] &= E[D \log(\pi(X'\gamma)/\pi(X'\gamma_0)) + (1 - D) \log((1 - \pi(X'\gamma))/(1 - \pi(X'\gamma_0))) | X] \\ &= \pi(X'\gamma_0) \log(\pi(X'\gamma)/\pi(X'\gamma_0)) + (1 - \pi(X'\gamma_0)) \log((1 - \pi(X'\gamma))/(1 - \pi(X'\gamma_0))) \\ &\leq 0. \end{aligned} \quad (\text{A.5})$$

The last inequality follows from $\log \lambda \leq \lambda - 1$, for $\lambda > 0$, which is strict for $\lambda \neq 1$. Since $\pi(X'\gamma) \neq \pi(X'\gamma_0)$ with positive probability, then the inequality in equation (A.5) is strict with positive probability and $E[m_\gamma - m_{\gamma_0}] < 0$ for $\gamma \neq \gamma_0$. Since $\pi(v)$ is bounded away from zero and one in V , then the absolute value of m_γ is bounded by a constant